Computational Work 3

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Course: Computational Matrix Methods

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Abstract

This work investigates two least-squares problems using data derived from the student's enrollment number. First, an overdetermined linear system $A\mathbf{x} \approx \mathbf{b}$, with $A \in \mathbb{R}^{5\times 4}$, is solved using the QR factorization $A = QR$. The least-squares solution is obtained by solving the resulting upper triangular system. The existence of a unique solution is discussed in terms of the rank of the coefficient matrix.

Second, a quadratic polynomial is fitted to six experimental data points using the least-squares formulation. The associated design matrix is constructed, and the polynomial coefficients are determined computationally. The quality of the fit is assessed by plotting the experimental data and the fitted curve and calculating the mean squared error.

The numerical implementation explicitly constructs the principal linear algebra algorithms, avoiding high-level routines for QR factorization and least-squares solutions. This approach connects the mathematical foundations of least squares with their computational implementation and numerical evaluation.

Keywords: QR factorization; least squares; quadratic polynomial fitting; numerical linear algebra.

References

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